EFFECTIVENESS OF ETALE DESCENT DATA REDUCES TO AFFINE OPENS OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original equation 0BTI: sources/stacks-chow/tags.txt, source lines 14841–14841. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02VR: sources/stacks-chow/tags.txt, source lines 3778–3778. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02VY: sources/stacks-chow/tags.txt, source lines 3785–3785. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let (V, φ) be a descent datum relative to X/S with $V \to X$ étale. Let $S = \bigcup S_i$ be an open covering. Assume that*

1. *the pullback of the descent datum (V, φ) to $X \times_S S_i/S_i$ is effective,*
2. *the functor (0BTI) for $X \times_S (S_i \cap S_j) \to (S_i \cap S_j)$ is fully faithful, and*
3. *the functor (0BTI) for $X \times_S (S_i \cap S_j \cap S_k) \to (S_i \cap S_j \cap S_k)$ is faithful.*

Then (V, φ) is effective.

Proof. (Recall that pullbacks of descent data are defined in Descent, Definition 02VR.) Set $X_i = X \times_S S_i$. Denote (V_i, φ_i) the pullback of (V, φ) to X_i/S_i. By assumption (1) we can find an étale morphism $U_i \to S_i$ which comes with an isomorphism $X_i \times_{S_i} U_i \to V_i$ compatible with *can* and φ_i. By assumption (2) we obtain isomorphisms $\psi_{ij} : U_i \times_{S_i} (S_i \cap S_j) \to U_j \times_{S_j} (S_i \cap S_j)$. By assumption (3) these isomorphisms satisfy the cocycle condition so that (U_i, ψ_{ij}) is a descend datum for the Zariski covering $\{S_i \to S\}$. Then Descent, Lemma 02VY (which is essentially just a reformulation of Schemes, Section 01JA) tells us that there exists a morphism of schemes $U \to S$ and isomorphisms $U \times_S S_i \to U_i$ compatible with ψ_{ij}. The isomorphisms $U \times_S S_i \to U_i$ determine corresponding isomorphisms $X_i \times_S U \to V_i$ which glue to a morphism $X \times_S U \to V$ compatible with the canonical descent datum and φ. □

Date: stacks-project:etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.